Language Learning Browser Extension

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Introduction

- Browser-based language learning extension designed to help with contextual and passive vocabulary learning while web browsing without interrupting reading flow
- Developed as a Chrome extension using Manifest V3, JavaScript, HTML, and CSS
- Scans webpage text automatically, replaces selected English words with French equivalents, and displays original English words through hover tooltips
- Shows potential for contextual vocabulary learning during normal browsing activities, but is limited by reliance on a static dictionary and doesn't include advanced language processing or grammatical analysis

Methods

- Injects a content script into webpages, scans webpage text nodes for predefined English words, replaces selected English words with French equivalents using a static dictionary, and highlights replaced words using HTML <span> elements
- Hovering over a translated word displays the original English word, and its phonetic pronunciation
- Uses DOM manipulation to update webpage text without affecting webpage structure or functionality, and updates dynamically as the user scrolls through the page
- Extension tested for loading times and translation accuracy

Discussion

Results:

- Original design was met because the result is a functional browser extension that replaces webpage words while scrolling, maintains webpage structure and functionality, and displays hover pop-ups with definitions
- Achieved 100% replacement accuracy in controlled testing
- Functions as a simplified version of commercial contextual learning tools, which could be adapted for accessibility support and developing English vocabulary

Applications:

- Reinforcing vocabulary learned in an academic setting
- Self-study language learning
- Passive vocabulary retention

Future Research:

- Optimal number of word substitutions for learning vs. comprehension
- Impact of adaptive difficulty on long-term recall
- Effects on grammatical understanding
- Compare contextual learning to traditional flashcard methods

Findings

Product Performance:

- Fully functional language-learning browser extension prototype
- Successfully injected a content script into webpages, replaced English words with French equivalents using a static dictionary, and displayed original English words and phonetic spelling when hovered over
- Replacements only occurred for whole words, which prevented substitutions for words inside longer words
- Did not modify restricted webpage elements and maintained webpage structure and interface performance

Testing Results:

- Tested on a local webpage containing all 10 dictionary words and compared expected substitutions to actual substitutions
- Results:
  - There was a 100% replacement accuracy and all expected substitutions were completed successfully
  - there was no observable delay in loading the webpage and the extension operated smoothly alongside the content of the webpage

Areas for Additional Testing:

- Scalability can be tested on webpages with high text volume
- Qualitative testing can measure the readability, reading fluency, and distraction levels for the user

Future Research

- Implement translation API support
- Add dynamic dictionaries based on user fluency level
- Randomize word replacement frequency to improve readability
- Improve handling of plural and alternate word forms

Limitations

- Plural words (ex: “dogs”) were not replaced unless manually added to the dictionary
- Scalability testing on larger, text-heavy webpages is still needed
- More qualitative testing required for readability, reading fluency, and user distraction level

References

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Leonardo English. (2025, April 14). *Active vs. Passive Listening: Why You're Not Improving Your English.* YouTube. https://youtu.be/fG5aNNP0FJE

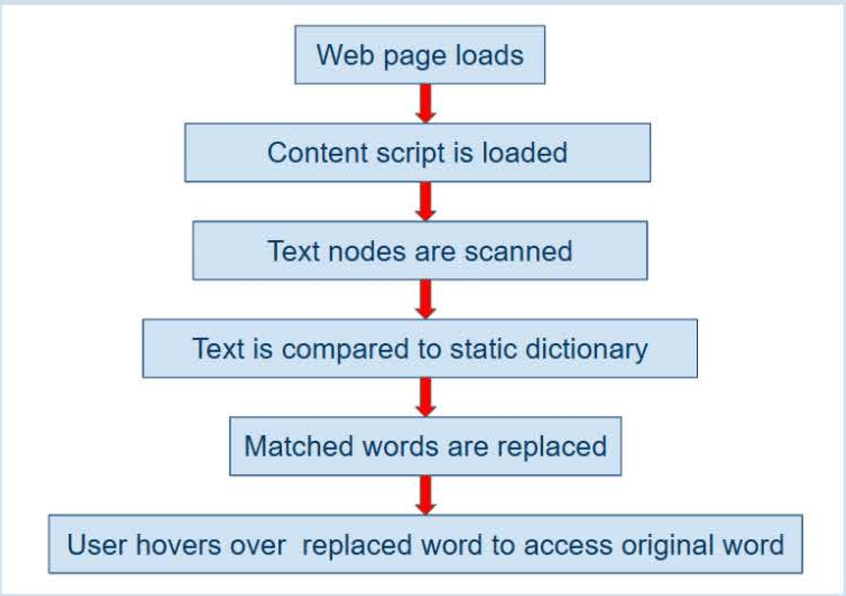

Figure 1. Flow chart showing how the code works

Figure 2. Initial Mock-Up

Figure 3. Product demonstration